APEXPLANET INTERNSHIP

TASK 1: DATA IMMERSION & WRANGLING

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Course: Bachelor of Computer Applications (BCA)

Internship Task: Task 1 – Data Immersion & Wrangling

Tools Used: Python, Pandas, NumPy, Jupyter Notebook

Project: Sales Dataset Cleaning and Preparation

1. Introduction

Data cleaning and preparation are important steps in the data analysis process. Real-world datasets may contain missing values, duplicate records, inconsistent data types, and unusual numerical values. These issues can affect the accuracy of analysis and business decisions.

This project focuses on understanding, assessing, cleaning, and preparing a sales dataset using Python and its data analysis libraries. The dataset contains order details, customer information, product categories, quantities, unit prices, and total sales.

The main goal was to transform the raw dataset into a clean, consistent, and analysis-ready dataset.

2. Objectives

The objectives of this task were:

- To understand the structure and contents of the dataset.
- To create a data dictionary describing the dataset columns.
- To assess missing values and duplicate records.
- To identify inconsistent data types and potential outliers.
- To clean and transform the data using Pandas.
- To validate sales calculations.
- To export the final cleaned dataset in CSV format.

3. Tools and Technologies

Python: Used to implement the data cleaning process.

Pandas: Used to load, inspect, clean, transform, and export the dataset.

NumPy: Used for numerical validation of sales calculations.

Jupyter Notebook: Used to execute Python code and document the analysis process.

CSV: Used to store the cleaned dataset.

4. Dataset Description

The dataset represents sales transactions and contains 1,000 rows and 12 columns.

The columns include order identifiers, order dates, customer information, product details, quantities, unit prices, and total sales values.

4.1 Data Dictionary

Column	Description
Order_ID	Identifier assigned to an order
Order_Date	Date on which the order was placed
Customer_ID	Identifier assigned to a customer
Customer_Name	Customer name or label in the dataset
Age	Customer age
Gender	Gender value recorded in the dataset
City	Customer city
Product	Purchased product
Category	Product category
Quantity	Number of units purchased
Unit_Price	Price per unit
Total_Sales	Total sales amount recorded for the transaction

5. Data Quality Assessment

The dataset was inspected using Pandas functions such as `head()`, `info()`, `dtypes`, `isnull().sum()`, `duplicated().sum()`, and `value_counts()`.

5.1 Data Types

Initially, the `Order_Date` column was stored as an object (text) data type. It was converted to the datetime data type to support date-based analysis.

The numeric columns included `Age`, `Quantity`, `Unit_Price`, and `Total_Sales`. Identifier and descriptive columns were retained as text or categorical values as appropriate.

5.2 Missing Values

The initial assessment identified the following missing values:

Column	Missing Values	Percentage
Age	20	2.0%
City	13	1.3%
Other columns	0	0%

The missing values in `Age` were filled using the median age. The missing values in `City` were replaced with the label `Unknown`.

After cleaning, all 12 columns contained zero missing values.

5.3 Duplicate Records

The dataset was checked for complete duplicate rows using `df.duplicated().sum()`.

The assessment found zero complete duplicate rows. Therefore, no complete duplicate records needed to be removed.

5.4 Categorical Data Assessment

The categorical columns, including `Gender`, `City`, `Product`, and `Category`, were examined using frequency counts. The displayed values did not reveal obvious capitalization or spelling inconsistencies requiring correction.

Missing city values were handled using the `Unknown` label.

6. Data Cleaning and Transformation

The following operations were performed:

1. **Date conversion:** Converted `Order_Date` to datetime format using Pandas.
2. **Missing age values:** Filled missing `Age` values with the median.
3. **Missing city values:** Replaced missing `City` values with `Unknown`.
4. **Duplicate check:** Checked for complete duplicate rows.
5. **Numerical validation:** Checked numerical columns for values less than or equal to zero.

6. **Outlier detection:** Applied the Interquartile Range (IQR) method to numerical columns.
7. **Sales validation:** Compared recorded sales values with calculated sales values.
8. **Export:** Saved the cleaned dataset as `sales_data_cleaned.csv`.

7. Outlier Detection

The IQR method was used to identify potential outliers.

The interquartile range is calculated as:

IQR = Q3 − Q1

The lower and upper boundaries are:

- Lower limit = Q1 − 1.5 × IQR
- Upper limit = Q3 + 1.5 × IQR

7.1 Results

Column	Potential Outliers
Age	0
Quantity	0
Unit_Price	0
Total_Sales	19

The IQR method identified 19 potential outliers in `Total_Sales`. No potential outliers were identified in the other three numerical columns.

These 19 records were flagged for investigation rather than automatically deleted. Extreme sales values may represent genuine transactions, so further business-context checks would be needed before deciding whether any records should be corrected or removed.

8. Numerical and Sales Validation

The numerical columns were checked for values less than or equal to zero.

Column	Non-positive Values
Age	0
Quantity	0
Unit_Price	0
Total_Sales	0

The sales calculation was validated using the formula:

Calculated Total Sales = Quantity × Unit_Price

The recorded `Total_Sales` values were compared with calculated values using NumPy's `isclose()` function with a small numerical tolerance.

Result: 0 sales calculation mismatches.

All records passed this sales consistency check within the specified tolerance.

9. Final Dataset

After the cleaning and validation steps, the final dataset had the following characteristics:

Metric	Final Result
Total rows	1,000
Total columns	12
Missing values	0
Complete duplicate rows	0
Sales calculation mismatches	0
Potential `Total_Sales` outliers	19
Non-positive values in checked numerical columns	0

The final dataset was exported as `sales_data_cleaned.csv`, with the DataFrame index excluded from the output.

10. Deliverables

The project deliverables include:

1. **Cleaned dataset:** `sales_data_cleaned.csv`
2. **Data dictionary:** `data_dictionary.md`
3. **Jupyter Notebook:** `ApexPlanet_Task1_Data_Wrangling.ipynb`
4. **Final report:** `ApexPlanet_Task1_Final_Report.docx`
5. **GitHub repository:** A repository containing the notebook, data dictionary, cleaned dataset, and project README.

11. Learning Outcomes

Through this task, I gained practical experience in:

- Inspecting datasets using Pandas.
- Understanding and converting data types.
- Identifying and handling missing values.
- Checking for duplicate records.

- Examining categorical data for inconsistencies.
- Detecting potential outliers using the IQR method.
- Validating numerical calculations using NumPy.
- Exporting cleaned data into CSV format.
- Documenting a data cleaning workflow for reproducibility.

12. Conclusion

This task provided practical experience in the initial stages of a data analysis project. The sales dataset was inspected, cleaned, validated, and exported into an analysis-ready CSV file.

The final dataset contains 1,000 rows and 12 columns, with no missing values, no complete duplicate rows, and no sales calculation mismatches. Nineteen potential outliers were identified in `Total_Sales` and retained pending further investigation.

The project demonstrates the importance of data quality assessment, careful data cleaning, validation, and documentation before performing further analysis or visualization.

Submitted as part of the ApexPlanet Internship – Task 1: Data Immersion & Wrangling.